

# Baseline Experiments and Initial Failure Analysis

Progress update: synthetic task baselines, validation results, and next steps

Length generalisation in Transformer sequence models

# We have completed the baseline pass

The project is now between baseline experiments and failure analysis

## What is done

Four synthetic tasks are implemented, generated, trained, and evaluated with a learned-position baseline.

## What is learned

Delayed copy and reverse show clean train-short/test-long failure. Addition needs shorter training length.

## What is not ready

Associative recall remains weak even after easier diagnostics, so it should not be scaled yet.

- The next stage is positional encoding comparison on validated tasks.

# The current task set covers memory, order, arithmetic, and retrieval

All tasks follow the train-short/test-long protocol up to 1024

## Delayed copy

Recall an earlier sequence after a separator.

## Reverse

Output tokens in the opposite order.

## Addition

Digit-wise arithmetic with carry propagation.

## Associative recall

Retrieve a value paired with a queried key.

## Baseline setup

- Transformer encoder baseline with learned absolute positional embeddings
- Primary maximum test length: 1024
- Metrics: token accuracy and exact-match accuracy

# Two tasks show clean length-generalisation failure

The model solves the training length, then collapses on longer unseen lengths

| Task         | Train | Test 128   | Test 256 | Test 1024 | Interpretation                           |
|--------------|-------|------------|----------|-----------|------------------------------------------|
| Delayed copy | 128   | 100% exact | 0% exact | 0% exact  | Learns short length; fails extrapolation |
| Reverse      | 128   | 100% exact | 0% exact | 0% exact  | Learns short length; fails extrapolation |

## Main takeaway

Delayed copy and reverse are ready for positional-encoding comparison because the baseline has learned the training distribution but fails beyond it.

# Addition works only after reducing the training length

This separates task learnability from length extrapolation

| Setup      | Train length | Exact@train | Exact@next | Token@1024 | Meaning                           |
|------------|--------------|-------------|------------|------------|-----------------------------------|
| Initial    | 128          | 0%          | 0%         | 10.1%      | Too hard at 128                   |
| Validation | 16           | 94.6%       | 0%         | 10.1%      | Learns short length; fails longer |

- For addition, train length 16 is the validated baseline for PE comparison.
- Train length 128 is useful evidence of training-length difficulty, not extrapolation.

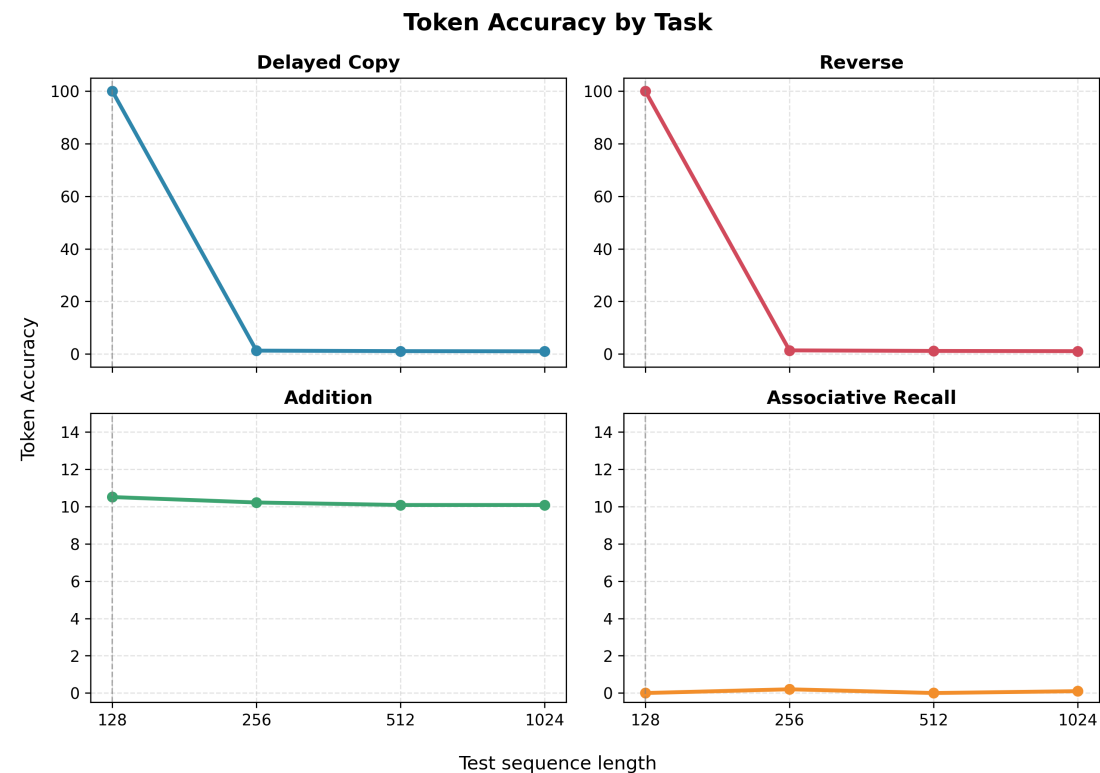
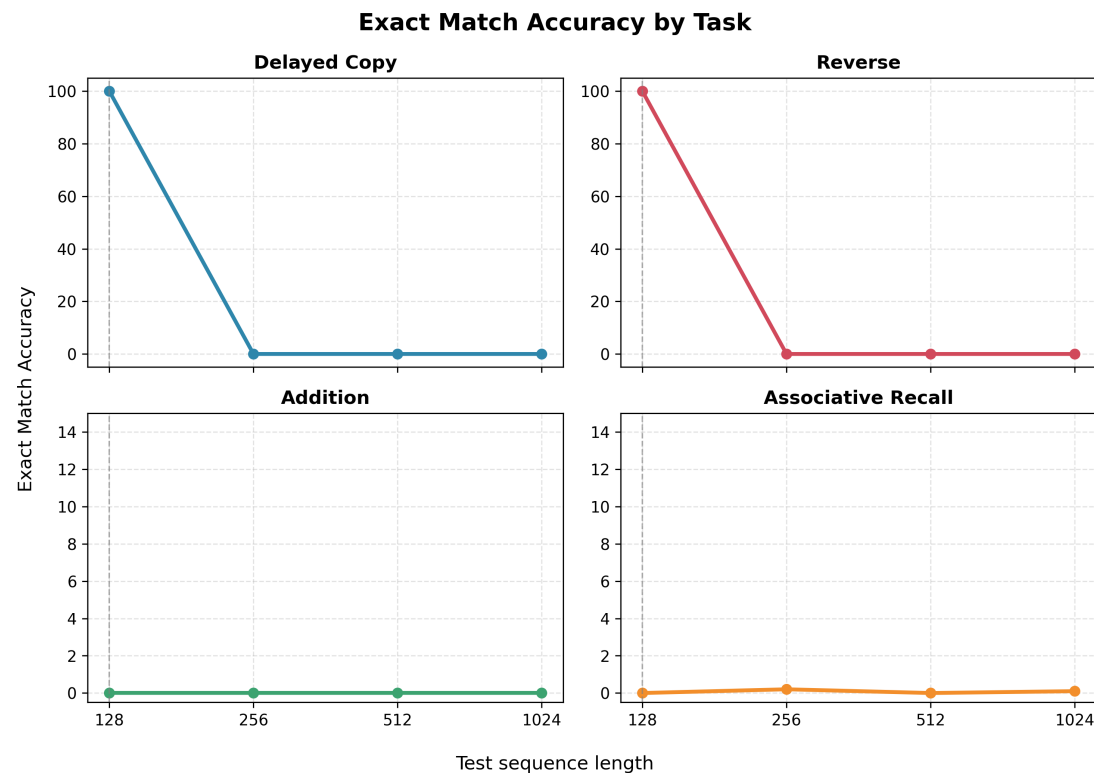
# Associative recall is still a task-design issue

The model remains near random after small-vocabulary and shorter-length diagnostics

| Diagnostic  | Train length | Exact@train | Exact@longest | Conclusion         |
|-------------|--------------|-------------|---------------|--------------------|
| Original    | 128          | 0%          | 0.1%          | Not learned        |
| Small vocab | 16           | 2.3%        | 0.3%          | Still near random  |
| Debug       | 8            | 9.3%        | 0.4%          | Weak learning only |

# The plots now support the first failure taxonomy

Baseline results already split into distinct failure types



**Pattern A:** learned at train length, failed on longer lengths.

**Pattern B:** did not reliably learn the training length.

# Next step: positional encodings on validated tasks

The baseline is ready to extend, but only where the task is learnable

| Priority | Run next                           | Reason                                             |
|----------|------------------------------------|----------------------------------------------------|
| 1        | Learned vs sinusoidal vs NoPE      | Simplest PE comparison; low implementation risk    |
| 2        | Delayed copy, reverse, addition-16 | These tasks have validated baseline behaviour      |
| 3        | Add ALiBi and RoPE                 | Core paper comparison once the model API is stable |
| 4        | Multiple seeds                     | Move toward robustness after PE pilot              |

**feedback**

Does this validation plan justify moving to positional encodings on the three validated tasks while associative recall remains under task-design review?